

Impact Statement

The SuperBPE Tokenizer for African Languages aims to improve how African languages are represented in language-model technologies by developing a tokenizer designed around the continent's linguistic, social, cultural, and economic contexts. Tokenization is an early stage of language processing, but its effects can propagate into machine translation, speech and text systems, information retrieval, educational tools, and generative AI. Better representation at this foundational level could make AI tools more accessible and useful for African language speakers, developers, translators, researchers, and educators.

The potential linguistic benefit is greater efficiency and preservation of meaning across diverse language structures. African languages are not linguistically uniform: they include tonal languages, languages with extensive morphology and affixation, and languages with different writing systems, orthographic conventions, and diacritic use. For example, Yoruba uses tone and diacritics that can distinguish meaning, while many Bantu languages express grammatical information through prefixes and other morphological patterns. Similar considerations arise across languages such as Igbo, Hausa, Amharic, Wolof, Zulu, Xhosa, and Swahili, among many others. A tokenizer trained on diverse African language corpora should therefore be evaluated not only on average token counts, but also on whether it preserves meaningful linguistic information across different language families and writing conventions.

Socially and culturally, the project could support greater participation of African communities in the development and use of language technologies. More effective tokenization could help enable local communication, education, translation, and access to digital information, while supporting the continued digital use of languages that are poorly represented in existing technologies. This connects to values of inclusivity, fairness, dignity, autonomy, cultural preservation, and community well-being. It also reflects Ubuntu and related ideas of relationality and collective flourishing: AI should strengthen people's ability to participate and benefit together, rather than treating communities and languages merely as sources of data.

There are, however, important risks. A tokenizer covering many African languages could still favour languages with larger datasets or greater online representation, leaving lower-resource languages behind. It may also inadequately handle tone, diacritics, morphology, orthographic variation,

cultural expressions, or language-specific conventions. Poor representation could therefore reproduce existing inequalities or weaken the quality of downstream systems. Community-informed evaluation and language-specific testing are important for identifying these harms, and transparency about data coverage, limitations, and performance differences is necessary.

The project also considers economic, computational, and environmental consequences. Reducing unnecessary tokens lowers inference costs, memory requirements, latency, and computational demand, which is particularly valuable where access to computing resources is constrained. Overall, the aim is to create a reusable and transparent tool that improves efficiency without sacrificing linguistic diversity, fairness, or community values. Because the relationship between better tokenization and real-world model quality remains uncertain, the project should continue testing across languages and downstream tasks, using evidence and community feedback to increase benefits and reduce potential harms.